

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech/ Ph.D (Autumn Semester End Exam Dec. 2022)

Subject : Statistical Computing for Data Science

Course Code : AM 604D

Duration : 03 Hrs.

Max. Marks : 50

- Marks are indicated against each question.
- Write clearly and legibly. All diagrams to be neatly labelled.
- Calculators and Statistical tables are allowed.**
- Before attempting please ensure that you have the correct question paper

- Let X_1, X_2 be two random variables with covariance matrix $\begin{bmatrix} 9 & \sqrt{6} \\ \sqrt{6} & 4 \end{bmatrix}$. Obtain the principal components and find the percentage of total error explained by each component. [10]
- Given the following set of data find out the multiple regression plane and check whether the regression is significant at 0.05 significance level. [10]

Y	10.98	7.82	8.11	10.36
X ₁	35.3	70.7	70	33.4
X ₂	20	21	22	20

- Discuss in detail about Linear and Non-Linear regression models with examples. [6]
 - Explain about Logistic regression and Poisson Regression. [4]
- State central Limit theorem. [2]
 - Find the probability of getting more than 25 sevens in 100 tosses of a pair of fair dice. [3]
 - Find the 99% confidence intervals for the variance and variance ratios of normally distributed populations. [5]
- The mean height of 50 male students who showed above average participation in college athletics was 68.2 inches with a standard deviation of 2.5 inches, while 50 male students who showed no interest in such participation had a mean height of 67.5 inches with a standard deviation of 2.8 inches. Test the hypothesis that male students who participate in college athletics are taller than other male students. Also find P value of the test. 0.05 [10]

$P > \alpha$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
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M.Tech/MS/PhD Examinations

Assessment – I - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 25 – 02 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. Before attempting please ensure you have the correct question paper.

Q1	In python is there any difference between the following: $a=123456789$ and $b= 1.23456789*10**8$ explain with justification. What will be the output if both variables a and b are printed?	[2.5]
Q2	ECG is the acronym for the electrocardiogram, is a sequence that always starts with the numbers 1 and 2. In this sequence, every number that succeeds is the smallest (not already present in the sequence) and shares a factor (excluding 1) with its preceding number. Every number in the ECG Sequence (besides 1 and 2) has a different index from its natural index in a normal numeric sequence. See the example below to establish the ECG Sequence Index of number 3. <code>SEQUENCE = [1, 2]</code> 3 = no factors shared with 2 4 = shares factor 2 with number 2 <code>SEQUENCE = [1, 2, 4]</code> 3 = no factors shared with 4 5 = no factors shared with 4 6 = shares factor 2 with number 4 <code>SEQUENCE = [1, 2, 4, 6]</code> 3 = shares factor 3 with number 6 <code>SEQUENCE = [1, 2, 4, 6, 3]</code> Number 3 is at index 4 in the ECG Sequence. Given an integer n implement a function that returns its ECG Sequence Index. Examples: <code>ecg_seq_index(3) → 4</code>	[2.5]

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

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M.Tech/MS/PhD Examinations

Assessment – II - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 23 – 03 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions.
Calculators are allowed.
2. Before attempting, please ensure you have the correct question paper.

Q1	Derive Normal Equation for Linear Regression.	[2.5]
Q2	Suppose a sensor generates a sequence of numbers between 1 and 10 for each reading, WAP to compute the frequency of all values for each reading. Example: For six readings the array is: <pre>array([[9, 9, 4, 8, 8, 1, 5, 3, 6, 3], [3, 3, 2, 1, 9, 5, 1, 10, 7, 3], [5, 2, 6, 4, 5, 5, 4, 8, 2, 2], [8, 8, 1, 3, 10, 10, 4, 3, 6, 9], [2, 1, 8, 7, 3, 1, 9, 3, 6, 2], [9, 2, 6, 5, 3, 9, 4, 6, 1, 10]])</pre> Then the output should be: <pre>[[1, 0, 2, 1, 1, 1, 0, 2, 2, 0], [2, 1, 3, 0, 1, 0, 1, 0, 1, 1], [0, 3, 0, 2, 3, 1, 0, 1, 0, 0], [1, 0, 2, 1, 0, 1, 0, 2, 1, 2], [2, 2, 2, 0, 0, 1, 1, 1, 1, 0], [1, 1, 1, 1, 1, 2, 0, 0, 2, 1]]</pre>	[2.5]
Q3	Access the effects of outliers while training a linear regression with the help of an example.	[2.5]
Q4	How can you verify that gradient descent has converged at a global minimum for convex and non-convex error functions?	[2.5]

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ALL THE BEST

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M.Tech/MS/PhD Examinations

Assessment – III - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 27 – 04 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions.
Calculators are allowed.
2. Before attempting, please ensure you have the correct question paper.

Q1	Prove that the derivative of the sigmoid function is a function of the sigmoid function. <i>Qn't</i>	[2.5]
Q2	Show that a simple perceptron model generate a nonlinear decision boundary by giving some example.	[2.5]
Q3	Find the SVD of the matrix $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$	[2.5]
Q4	Given the vector $\mathbf{x} = [1, 1, 1]^T$, which vector $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ "best lies in the direction of $\mathbf{x}$ " (in the sense of the subspace spanned by the vector, as in the OLS).	[2.5]

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ALL THE BEST

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

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M.Tech/MS/PhD Examinations

Re-Assessment – II - (Spring 2023)

Course Name: Data Science Tools & Techniques

Course Code: AM624D

Date: 13 – 04 - 2023

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly Attempt all questions. Calculators are allowed.
2. Before attempting, please ensure you have the correct question paper.

Q1	In class, we derived a gradient descent learning algorithm for simple linear regression where we assumed: $y = w_0 + w_1x_1 + \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, \sigma^2)$ Derive the same for : $y = w_0 + w_1x_1 + w_2x_2 + w_3x_1^2 + \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, \sigma^2)$ where your learning algorithm will estimate the parameters $w_0, w_1, w_2,$ and w_3. Write down an expression for $P(y x_1, x_2)$ Calculate the gradient with respect to all four parameters in w.	[05]
Q2	Suppose a sensor generates a sequence of numbers between 1 and 10 for each reading, WAP to compute the frequency of all values for each reading. Example: For six readings the array is: <pre>array([[9, 9, 4, 8, 8, 1, 5, 3, 6, 3], [3, 3, 2, 1, 9, 5, 1, 10, 7, 3], [5, 2, 6, 4, 5, 5, 4, 8, 2, 2], [8, 8, 1, 3, 10, 10, 4, 3, 6, 9], [2, 1, 8, 7, 3, 1, 9, 3, 6, 2], [9, 2, 6, 5, 3, 9, 4, 6, 1, 10]])</pre> Then the output should be: <pre>[[1, 0, 2, 1, 1, 1, 0, 2, 2, 0], [2, 1, 3, 0, 1, 0, 1, 0, 1, 1], [0, 3, 0, 2, 3, 1, 0, 1, 0, 0], [1, 0, 2, 1, 0, 1, 0, 2, 1, 2], [2, 2, 2, 0, 0, 1, 1, 1, 1, 0], [1, 1, 1, 1, 1, 2, 0, 0, 2, 1]]</pre>	[2.5]
Q3	Explain with examples of different types of sampling.	[2.5]

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Defence Institute of Advanced Technology, Pune

Course Name: AM 607, Assessment-2, Answer All questions.

1. Explain the logistic regression with mathematical expressions. If the decision boundary learnt by the logistic regression takes the form $f(x,y,z) = 3x + 2y - 7z + 10$, then name the following samples which belong to the same category. $A=[1,1, 3]$, $B= [1,1,1]$, $C=[5,0,1]$, $D=[0,0,1]$
2. Explain or Justify the scenario where Accuracy is unreliable as a performance metric and what is the alternative metric that addresses the shortcomings of Accuracy.
3. Explain what is the impact of noise or outliers in learning decision trees, and how it can be addressed
4. True or False: (a). KNN cannot be used for regression. (b). Logistic regression is used for regression (c). Which classification algorithm works better when features are categorical? (d). The performance of KNN does not depend on the similarity measure. (e). Decision Trees cannot be built for multi-label classification

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
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M.Tech/MS/PhD Examinations
Assessment – I - (Autumn 2022)

Course Name: Data Structures and Algorithms with C

Course Code: AM607D

Date: 14 – 09 - 2022

Duration: 01 Hr

Max Marks: 10

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams (if any) to be neatly labeled. Attempt all questions.
2. Before attempting, please ensure you have the correct question paper.

Q1	Write any two differences between structural and object-oriented languages.	[2]
Q2	Explain the different data types with the size and ranges.	[2]
Q3	a. Explain with an example of while and do-while loop. b. Write the while loop of the given for loop code. <pre>for (i=0; i<10; i++) sum+=i;</pre>	[1+1]
Q4	a. Write a C program to print the given pattern. <pre> * * * * * * * * * * * * * * * *</pre> b. Write a C program to reverse a given number.	[2+2]

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$a = 5, b = 10$

$a = a - b$
 $= 5 - 10$
 $b = 5$
 $a = b + b$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
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M.Tech / Ph.D Assessment-I (26th Sept. 2022)

Course (Subject) : **Advanced Optimization Techniques** Course Code : **AM-603 /**
Computer Oriented Optimization Techniques **AM-603D**

Duration : **01 hr**

Max Marks : **10**

Note. Attempt **all** questions from the following

1. a). Write the Phases of O.R. Model.
b). Write the Simplex method algorithm.
2. Use Big-M method to solve the following L.P.P

$$\text{Minimize } Z = 2x_1 + x_2$$

$$\text{S.to. } 3x_1 + x_2 = 3$$

$$4x_1 + 3x_2 \geq 6$$

$$x_1 + 2x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

3. Use Two-Phase method to solve the following L.P.P

$$\text{Minimize } Z = x_1 - 2x_2 - 3x_3$$

$$\text{S.to. } -2x_1 + x_2 + 3x_3 = 2$$

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$

$$R_2 - 4R_1 \quad \text{S) } 6(1.2) \quad \text{S) } 9(1.8)$$

$$R_3 - R_1 \quad \frac{5}{10} \quad \frac{5}{40}$$

$$3) 20(0.6)$$

$$4) 10(0.2)$$

$$R_1 - \frac{1}{3}R_2 \quad -\frac{2}{3} + \frac{3}{1}$$

$$\frac{1}{3} - \frac{1}{3}$$

$$1 - \frac{1}{3}$$

$$1 - \frac{1}{3} \left(\frac{2}{3} \right)$$

$$1 - 2 \left(\frac{1}{3} \right)$$

$$1 - \frac{1}{3} \left(\frac{2}{3} \right)$$

$$-\frac{1}{3} \left(-\frac{2}{3} \right)$$

$$\frac{5}{3} - \frac{1}{3}$$

$$R_2 - \frac{1}{3}R_2$$

$$3 - \frac{10}{3} \left(\frac{2}{3} \right)$$

$$- \frac{1}{3} \left(-\frac{2}{3} \right)$$

$$3 - \frac{4}{3}$$

$$-1$$

$$2 - \frac{1}{3}$$

$$4 - 1$$

$$\frac{3}{1} \times \frac{3}{5}$$

$$9 \frac{1}{5}$$

$$1$$

$$2 \frac{1}{1} \times \frac{2}{5}$$

$$-\frac{2}{3} - \frac{1}{3} + 1$$

$$-\frac{1}{3} + \frac{1}{3}$$

$$3) 20(0.6)$$

$$4) 10(0.2)$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed to be University)
M.Tech /Ph.D END SEMESTER EXAMINATIONS : Nov. / Dec. 2022

Course (Subject) : **Computer Oriented Optimization Techniques** Code **AM-603-D**
Duration : **03 hrs** Max Marks : **50**

Note: 1 Before attempting , please ensure that you have the correct question paper.

2. Attempt **all Five** questions

3. Students may request the invigilator for required Tables (if any).

1). Solve the following L.P.P. by Simplex method (10 M)

Maximize $Z = -x_1 + 3x_2 - 2x_3$

S.to. $3x_1 - x_2 + 2x_3 \leq 7$

$-2x_1 + 4x_2 \leq 12$

$-4x_1 + 3x_2 + 8x_3 \leq 10$

$x_1, x_2, x_3 \geq 0$

2). Using Kuhn-Tucker conditions to solve the following NLPP. (10 M)

Max $Z = -x_1^2 - x_2^2 - x_3^2 + 4x_1 + 6x_2$

S.to. $x_1 + x_2 \leq 2$

$2x_1 + 3x_2 \leq 12$

$x_1, x_2 \geq 0$

3. a). Find the extreme points and their nature of the function (5+5 = 10 M)

$f(x_1, x_2) = x_1^3 + x_2^3 + 2x_1^2 + 4x_2^2 + 6$

b). Write a short note on Genetic Algorithm and its applications.

4.. Use Steepest Decent Method to solve the following problem. (10 M)

Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ starting from the point $X_1 = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$

5. A small project is composed of seven activities whose time estimates are listed in the table as follows: (10 M)

Activity		1-2	1-3	1-4	2-5	3-5	4-6	5-6
time estimates in weeks	t_o	5	1	2	3	1	2	1
	t_m	6	1	4	6	1	2	4
	t_p	7	7	12	15	1	8	7

a). Draw the Project net work

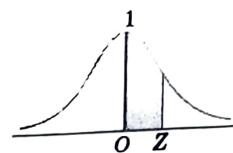
b). Find the critical path of the project and project duration.

c). What is the probability that the project will be completed at least 4 weeks earlier than expected. duration.

----- @@@@ -----

Handwritten calculations:
 $-\frac{4}{5} + \frac{6}{5} + 2 = 5.4$
 $\frac{2}{5}$
 0.31
 $-\frac{2}{5} + \frac{3}{5} - \frac{4}{5} + \frac{6}{5} + 2 = 2.4$
 $\frac{2}{5} + 2 = 2.4$
 1.75

Table III : Area under the Normal curve



z	0	1	2	3	4	5	6	7	8	9
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0754
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2258	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4782	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

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M.Tech EXAMINATIONS

2nd Evaluation EXAMINATION: Autumn 2022

Course: M.Tech. SUBJECT : Intelligent Algorithms Code CE615A

Duration: 01 Hrs.

Max Marks – 10

Note –

- 1.** Marks are indicated against each question.
- 2.** Write clearly and legibly. All diagrams to be neatly labelled.
- 3.** Answer all the questions.
- 4.** Please ensure that you have been issued the correct question paper.

Q1. Briefly explain the Minimum spanning tree with an example.

2 Mark

Q2. With the help of pseudo-code, briefly discuss the dynamic programming approach to Fibonacci number generation.

2 Mark

Q3. Briefly discuss the division method for Hash Function with an example.

2 Mark

Q4. With the help of pseudocode and diagram, briefly discuss the functions to find Minimum and Maximum in Binary Search Tree(BST).

2 Mark

Q5. Apply the greedy algorithm for the following Activity selection Problem and find the solution i.e., maximum-size subset of mutually compatible activities.

i	1	2	3	4	5	6	7	8	9	10	11
Si (start time)	1	3	0	5	3	5	6	8	8	2	12
Fi (Finish Time)	4	5	6	7	9	9	10	11	12	14	16

2 Mark

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
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M.Tech EXAMINATIONS

1st Evaluation EXAMINATION: Autumn 2022

Course: **M.Tech.** SUBJECT : **Intelligent Algorithms** Code **CE615A**

Duration: 01 Hrs.

Max Marks – 10

Note –

1. Marks are indicated against each question.
 2. Write clearly and legibly. All diagrams to be neatly labelled.
 3. Answer all the questions.
 4. Please ensure that you have been issued the correct question paper.
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Q1. With the help of Pseudocode, briefly discuss the Insertion Sort Algorithm.

2 Mark

Q2. Briefly discuss the designing of the algorithm with divide-and-conquer approach.

2 Mark

Q3. Briefly discuss the concept of loop invariant with three properties.

2 Mark

Q4. With the help of pseudocode, briefly explain the Merge-Sort algorithm.

2 Mark

Q5. A computer is having speed of 10 million instructions per second(MIPs). It must sort 10 million numbers. Compute the time it takes to sort the numbers using Insertion Sort and Merge sort. Assume that Insertion sort takes $2n^2$ instructions and merge-sort takes $50(n \lg_2 n)$ to sort n numbers

2 Mark

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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech EXAMINATIONS

End Semester EXAMINATION: Odd Semester 2022

Course: **M.Tech.** SUBJECT : **Intelligent Algorithms** Code **CE615A**

Duration: 03 Hr

Max Marks – 50

Note –

1. Marks are indicated against each question.
2. Write clearly and legibly. All diagrams to be neatly labelled.
3. Answer all the questions
4. Please ensure that you have been issued the correct question paper.
5. Scientific calculators are permitted.

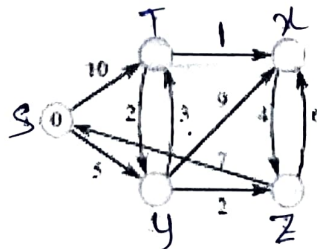
- Q1.** (a) Briefly discuss double hashing algorithm with the help of a diagram.
(b) Define and compare the three important asymptotic notations i.e. $O()$, $\Theta()$ and $\Omega()$
(c) With the help of a diagram, briefly discuss the recursion tree for the recurrence
 $T(n) = 3T(n/4) + cn^2$.

10 Marks

- Q2.** (a) Briefly discuss the Decision Tree Algorithm for classification listing the advantages and disadvantages of Decision Trees Algorithm.
(b) Briefly discuss the Merge-sort algorithm with the Pseudo-code.
(c) A computer programmer is required to sort 100 million numbers. She has two computers i.e., Computer A with speed of 10 billion instructions per second(10,000MIPS) and Computer B with speed of 10 million instructions per second(10MIPS). She has implemented program with Insertion sort Algorithm on Computer A which takes $2n^2$ instructions to sort n numbers. On Computer B, she has implemented program with Merge Sort Algorithm which takes $50n \log_2 n$ instructions to sort n numbers. Compare the time taken by both the computers to sort the input of 100 million numbers.

10 Marks

- Q3.** (a) Differentiate between Divide-and-conquer algorithms from Dynamic Programming.
(b) Differentiate between Breadth First Search (BFS) and Depth First Search(DFS).
(c) Using Dijkstra's shortest path algorithm, obtain the shortest path from source s for the following graph showing step by step analysis.



10 Marks
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DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University)

M.Tech EXAMINATIONS

3rd Evaluation EXAMINATION: Autumn 2022

Course: M.Tech. SUBJECT : Intelligent Algorithms Code CE615A

Duration: 01 Hrs.

Max Marks – 10

Note –

- 1. Marks are indicated against each question.**
 - 2. Write clearly and legibly. All diagrams to be neatly labelled.**
 - 3. Answer all the questions.**
 - 4. Please ensure that you have been issued the correct question paper.**
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Q1. Briefly discuss the Genetic Algorithm with the help of a flow chart.

2 Mark

Q2. Differentiate between Supervised, Unsupervised and Reinforcement Learning.

2 Mark

Q3. Briefly discuss the Lloyd's Algorithm for Clustering with the Pseudo-code.

2 Mark

Q4. With the help of Pseudo-code, briefly discuss the RSA Algorithm.

2 Mark

Q5. Draw the Confusion Matrix, compute the Classification Accuracy, Precision, Recall and F1-Score for the following data: TP = 560, TN = 330, FP = 60, FN = 50.

2 Mark

Subject : Scientific Computing

Course Code : AM 606D

Duration : 03 Hrs.

Max Marks : 50

- a. Marks are indicated against each question.
b. Before attempting please ensure that you have the correct question paper.

1. a) Find a linear mapping $F: R^3 \rightarrow R^3$ whose image is spanned by $(1,2,3)$ and $(4,5,6)$. [3]
 b) Find a linear mapping $G: R^4 \rightarrow R^3$ whose kernel is spanned by $(1,2,3,4)$ and $(0,1,1,1)$. [3]
 c) Find the homogeneous system whose solution space is spanned by $(1,1,2,1,1)$, $(1,2,1,4,3)$, $(3,5,4,9,7)$. [4]
2. a) Show that the Moore Penrose inverse is unique. Find the Moore Penrose inverse for $A = \begin{bmatrix} 1 & 2 & 0 \\ -2 & 1 & 2 \\ 1 & 3 & 1 \end{bmatrix}$ using QR decomposition. [10]
3. a) Discuss Overfitting and Underfitting for system of equations with examples. Also give the solution by writing appropriate Pseudo inverse in both the cases. [4]
 b) Find the largest eigen value and the corresponding eigen vector using power method for the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ [6]
4. a) Give the algorithm for solving system of equations using LU decomposition. [5]
 b) Discuss Gram Schmidt orthonormalization procedure. [5]
5. a) Using 3 point Gauss quadrature, evaluate $\int_0^1 \frac{dr}{1+r^2}$ [5]
 b) Discuss PCA in detail and give its application in Data mining with an example. [5]

$$\frac{9}{4} + 4 + \frac{25}{4}$$

$$\frac{9+16+25}{4}$$

$$\frac{50}{4}$$

2.034-0.011

$$2 \sqrt{50.4}$$

$$2 \sqrt{25.2}$$

M.Tech/MS/PhD END SEMESTER EXAMINATIONS: SPRING/AUTUMN-2022

Course Name: Artificial Algorithms and DSS

Course Code: CE 696A

Duration: 03Hrs

Q-Paper pages: 02

Max Marks: 50

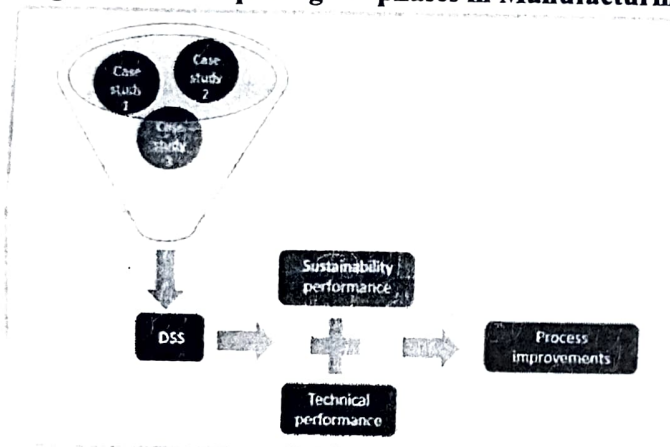
Date: June/July 2022

Note:

1. Write each QUESTION on new page, and Strictly maintain the sequence of ans.
2. Maintain the sequence of the answers written for sub questions in each QUESTION.
3. Marks are indicated against each QUESTION.
4. Write clearly & legibly. Neatly label & state the assumptions wherever necessary.
5. Before attempting, please ensure you have the correct question paper.
6. One Mark is reserved for the neatness & the ability to follow the above instructions.

QUESTION-I

1. State using block diagram the steps constituting the KDD process & State Noise in each step. [7 Marks]
2. Refer the figure given below depicting the phases in Manufacturing Sector.



2A. State Case Study 1, Case Study 2 and Case Study 3 as per the tabulated form given below- [3 Marks]

Case Study	Description	Nature of Data	Type of DSS	Justification
1				
2				
3				

2B. State any three measurable sustainability performance evaluation parameters applicable to the case study in 2A as per the tabulated form given below- [9 Marks]

Sr. No	Name of the parameter	Evaluation criteria/equation	Type of DSS	Justify
1				
2				
3				

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)

MTech/MS/PhD EXAMINATIONS

WRITTEN EXAMINATION – (SPRING/AUTUMN – 2022): Test-II

Course (Subject): AI and DSS

Code: CE-696A

Duration : 01 HOURS

Max Marks: 10

Date and Time: 17/10/2022 (11.00-12.00Hrs)

Note:

1. Answer **ALL** Questions.
 2. Assume suitable data, if necessary.
 3. Marks are indicated against each Question.
 4. Question paper consists of: 01 Page.
-

Q-1.

Explain perceptron and various activation functions.

[08]

Q-2.

Explain any learning rule used in case of neural networks.

[02]

.....

Instructor: Dr Sunita Dhavale,
Asst Professor CSE Department, DIAT, Pune
Email id:
sunitadhavale@diat.ac.in/sunitadhavale@gmail.com

Defence Institute of Advanced Technology

School of Computer Engineering & Mathematical Sciences

Artificial Intelligence & Decision Support Systems (CE696A)

MTech/PhD (2022-24)

Assessment 1 (30/09/2022)

Maximum Marks: 10

- Q.1 Describe the Turing test for artificial intelligence and justify its validity from a modern standpoint. [2.5]
- Q2. List problems for which the forward chaining inference technique is appropriate. Why backward chaining used for diagnostic problems? [2.5]
- Q3. What are the fundamental characteristics of an expert systems? What are the differences between expert systems and conventional programs? [2.5]
- Q4. The reliability of a particular skin test for tuberculosis (TB) is as follows: If the subject has TB, the test comes back positive 98% of the time. If the subject does not have TB, the test comes back negative 99% of the time. From a large population, in which 2 in every 10,000 people have TB, a person is selected at random and given the test, which comes back positive. What is the probability that the person actually has TB? [2.5]

.....End.....

1-0.99

~~0.00002~~

0.000198

0.0002

0.000196

0.000196 + 0.000196

0.000394

$P(P) = 0.98$
 $P(N) = 0.99$

$0.98 \times \frac{2}{10000}$

$0.98 \times \frac{2}{10,000} + 0.01 \times \frac{2}{10,000}$

0.000196

0.000196 +

0.000394

0.4974

0.000198

0.000002 + 0.000196

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY
(Deemed University)

MTech/MS/PhD EXAMINATIONS

WRITTEN EXAMINATION – (SPRING/AUTUMN – 2022): Test-II

Course (Subject): AI and DSS

Duration : 01 HOURS

Date and Time: 17/10/2022 (11.00-12.00Hrs)

Code: CE-696A

Max Marks: 10

Note:

1. Answer **ALL** Questions.
2. Assume suitable data, if necessary.
3. Marks are indicated against each Question.
4. Question paper consists of: 01 Page.

Q-1.

Explain perceptron and various activation functions.

[08]

Q-2.

Explain any learning rule used in case of neural networks.

[02]

.....

Instructor: Dr Sunita Dhavale,
Asst Professor CSE Department, DIAT, Pune
Email id:
sunitadhavale@diat.ac.in/sunitadhavale@gmail.com

Department of Computer Science & Engineering, DIAT

Deep Learning (CE631)
MTech/PhD

Assessment 1 (23/02/2023)
Maximum Marks: 10

Q1. Identify the significance of 'e' during the calculation of softmax activation [1]
a) To get output value between 0 and 1
b) To map output on real numbers
c) To avoid negative values
d) None of the above

Q2. If we use very high value for w in logistic function, then it resembles as.....? [1]

Q3. Define any 02 activation functions used in deep learning with graphical representations. Is it necessary to use differentiable activation functions ? Justify. [2]

Q4. How can Bias and Variance in the Error is associated with Overfitting and Underfitting. Explain diagramaticatly. [2]

Q5. Consider a discrete random variable **X** which can take one of 3 values from the set {x1, x2, x3}. A distribution over X defines the value of $\Pr(X = x) \forall x \in \{x1, x2, x3\}$. Consider two such distributions **P** and **Q** which are defined as follows, What will be the KL divergence between P and Q? (Use base-2 logarithm.) [2]

Probability Distribution	x1	x2	x3
P(X)	0	1	0
Q(X)	0.25	0.35	0.40

Q6. Differentiate any two Gradient Descent Methods with example. [2]

Department of Computer Science & Engineering, DIAT

Deep Learning (CE631)
MTech/PhD

Assessment 2 (22/03/2023)
Maximum Marks: 10

Q1. Given a 128x128x3 image and 6 filters of size 9x9x3, what will be the dimension of the output volume when the stride S=1 and the padding P=2? [1]

Q2. Does the size of the feature map always reduce upon applying the filters? Explain why or why not. [2]

Q3. Explain the significance of the RELU Activation function in Convolution Neural Network. [2]

Q4. Consider the Code Snippet shown in Fig. 1 and calculate the output of model.summary(). Here, we have no Non-Trainable Parameters. [5]

```
model = models.Sequential()

model.add(layers.Conv2D(32, (5,5), activation='relu', input_shape=(28, 28,1)))
model.add(layers.MaxPooling2D((2, 2)))

model.add(layers.Conv2D(64, (5, 5), activation='relu'))
model.add(layers.MaxPooling2D((2, 2)))

model.add(layers.Flatten())
model.add(layers.Dense(10, activation='softmax'))

model.summary()
```

Figure 1. Code Snippet

Department of Computer Science & Engineering, DIAT

Deep Learning (CE631)
MTech/PhD

Assessment 3 (01/05/2023)
Maximum Marks: 10

- Q1. Can you think of a few applications for a sequence-to-sequence RNN? What about a sequence-to-vector RNN? And a vector-to-sequence RNN? [2.5]
- Q2. Why do people use encoder-decoder RNN rather than plain sequence-to-sequence RNNs for automatic translation? [2.5]
- Q3. How could you combine a convolutional neural network with an RNN to classify videos? [2.5]
- Q4. Suppose you want to train a classifier and you have plenty of unlabeled training data, but only a few thousand labeled instances. How can autoencoders help? How would you proceed? [2.5]

Assessment 2 (06 May 2023, 1000-1100)

Course: Big Data Analytics (CE694) **Instructor:** Dr. Sushma Kumari, SoCE&MS, **Total marks:** 10marks

Questions

- Q1. Design Map and Reduce functions to take a very large files of integers and produce as an output. Write the complete flow of the MapReduce environment for the following
- The same set of integers, but with each integer appearing only once. **(1 mark)**
 - The count of the number of distinct integers in the output. **(1 mark)**
- Q2. There is an input file of 500 MB. How many splits HDFS will create by default and what would be the size of each split. **(1 mark)**
- Q3. Explain the architecture (using a diagram) and two main advantages of using HDFS. **(2 marks)**
- Q4. Find the eigenvalues, eigenvectors and find the SVD decomposition of the matrix $\begin{bmatrix} 1 & 2 \\ 5 & 4 \\ 3 & 5 \end{bmatrix}$. **(2 marks)**
- Q5. Write a working MySQL query for the following
- Create two tables "Students" with columns "Student_id", "Student_Name" and "Department". Further add the following values to the Students table: [101, Red Singh, MA], [102, Blue Babble, MA], [103, Green Gibbon, CSE] **(0.5 marks)**
 - Create another table "Officers" having columns "Officer_id", "Officer_Name" and "Rank". Further add the following values to the Students table: [101, Beige Brown, CSE], [102, Mango Miller, MA]. **(0.5 marks)**
 - Write a query to inner join the students and officers table. Also, write the output after joining the table. **(2 marks)**

Assessment 3 (19 May 2023, 1000-1100)

Course: Big Data Analytics (CE694) **Instructor:** Dr. Sushma Kumari, SoCE&MS, **Total marks:** 10marks

Questions

Q1. State True/False (0.5 *4 = 2 marks)

- (i) ☒ The predictors in the k-variable model identified by forward stepwise are a subset of the predictors in the (k+1)-variable model identified by forward stepwise selection.
- (ii) ☒ The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the (k + 1)- variable model identified by backward stepwise selection.
- (iii) ☐ The predictors in the k-variable model identified by backward stepwise are a subset of the predictors in the (k + 1)- variable model identified by forward stepwise selection.
- (iv) ☒ The predictors in the k-variable model identified by best subset are a subset of the predictors in the (k + 1)- variable model identified by best subset selection.

Q2. State two main difference between Ridge and LASSO regression. (1 mark)

Q3. Describe the four main assumptions of Linear Regression. (4 marks)

Q4. What is adjusted R^2 error? State its formula and meaning. (1 mark)

Q5. The table below provides a training data set containing 6 observations, 3 predictors, and 1 qualitative response variable. Suppose we wish to use this data set to make a prediction for Y when $X_1 = X_2 = X_3 = 0$ using K-nearest neighbors.

(a) Compute the Euclidean distance between each observation and the test point, $X_1 = X_2 = X_3 = 0$. (1 mark) (b) What is our prediction with K = 1? Why? (1 mark)

Observations	1	2	3	4	5
X_1	0	2	0	0	-1
X_2	3	0	1	1	0
X_3	0	0	3	2	1
Y	Red	Red	Red	Green	Red

Assessment 1 (27/02/2023)

Course: Big Data Analytics (CE694)

Instructor: Dr. Sushma Kumari, SoCE&MS, DIAT

Total marks: 10marks; marks distribution is indicated against each question

Questions

- Q1. Describe the four types of data analytics. **(2marks)**
- Q2. What is a CAP theorem? State its three principles. **(1mark)**
- Q3. Describe the ACID and BASE properties of a database. **(2marks)**
- Q4. Give any three characteristics of Big Data supported with an example. **(3marks)**
- Q5. Suppose there is a repository of ten million documents, and word w appears in 320 of them. In a particular document d , the maximum number of occurrences of a word is 15.
- a. What is the TF.IDF score for w if that word appears once? **(1mark)**
 - b. What is the TF.IDF score for w if that word appears five times? **(1mark)**

Subject: Machine Learning: Spring 2023

Assessment No. 1

Marks: 10

1. Considering a Hypothesis: $H_{\theta}(x) = \theta_0 + \theta_1 x$, with $\alpha=0.01$, $\theta_0 = \theta_1 = 0$, for the dataset. Suppose the values of (x^i, y^i) are: (1, 1), (2, 3), (3, 5), (4, 7), (5, 9). Using these values, what are the updated values of θ_0 and θ_1 after three iterations of gradient descent?
4M
2. Explain how stochastic gradient descent (SGD) differs from gradient descent. Give two advantages of SGD over gradient descent
3M
3. Explain the term Coefficient of Determination, and how the model performance is accessed based on it.
3M

$$\theta_1 = 0.88$$
$$\theta_0 = 0.1723$$

Assessment 3

Sub: Machine Learning: (AM623/AM623D)

~~~~~

1. For an SVM with a decision boundary of the form:  $W^T x + b = 0$ . The support vectors are  $x_1 =$ ,  $x_2 =$ ,  $x_3 =$ , ,  $y_1 = y_2 = -1$ ,  $y_3 = 1$ . Compute  $W$  and  $b$ .

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\alpha_1 = 0.2; \alpha_2 = 0.5 \quad \alpha_3 = 1 \quad y_3 = 1$$

(3 M)

2. Explain (a) : Under what circumstances do we choose Soft margin SVM?

(b) : How SVM can be used for Linearly non-separable data?

(2 M)

3. How to realize a Logistic regression classifier based on Neural Network (NN) (2.5 M)

4. Explain the difference in the NN architecture for Regression and Classification (2.5 M)

---

$$6.2x_1 + 7.9x_2 = -0.7$$

$$x_2 = -0.112$$
$$-0.08$$

DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY  
(Deemed University)  
M.Tech/MS/PhD Examinations  
END SEMESTER - Examination (Spring, 2023)

Course Name: Machine Learning

Course Code: AM 623/623D

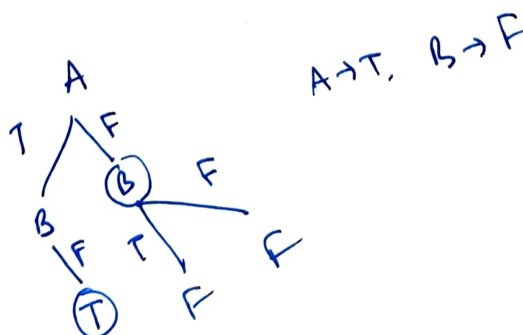
Duration : 03 Hours

Max Marks : 50

Note:

1. Marks are indicated against each question. Write clearly and legibly. All diagrams are to be neatly labelled. Calculators are allowed.
2. Attempt **ALL** questions.
3. Before attempting please ensure you have the correct question paper.

|            |           |                                                                                                                                                                                                                                                                                                                                                                                     |           |      |   |   |   |            |     |     |     |      |
|------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------|---|---|---|------------|-----|-----|-----|------|
| Q1         | a)        | Explain in detail what is needed for applying the machine learning algorithms to any given problem                                                                                                                                                                                                                                                                                  | 2M        |      |   |   |   |            |     |     |     |      |
|            | b)        | With respect to Regression- explain the terms overfitting, underfitting, and the right fit of the model with diagrams. Explain any two techniques that can address the problem of underfitting.                                                                                                                                                                                     | 5M        |      |   |   |   |            |     |     |     |      |
|            | c)        | What is feature normalization and why it is needed? How the Data: [10, 7, 15, 74, 99, 25] transforms after applying mean normalization                                                                                                                                                                                                                                              | 3M        |      |   |   |   |            |     |     |     |      |
| Q2         | a)        | Explain Polynomial regression in detail, and on what basis, Polynomial regression will be chosen over linear regression                                                                                                                                                                                                                                                             | 5M        |      |   |   |   |            |     |     |     |      |
|            | b)        | Explain the Coefficient of Determination ( $R^2$ ), and compute the $R^2$ value for Hypothesis: $h(x)=2x+3$ , considering the below data.<br><table border="1"><tr><td>x (input)</td><td>1</td><td>2</td><td>3</td><td>5</td></tr><tr><td>y (output)</td><td>4.8</td><td>7.2</td><td>8.6</td><td>12.8</td></tr></table>                                                             | x (input) | 1    | 2 | 3 | 5 | y (output) | 4.8 | 7.2 | 8.6 | 12.8 |
| x (input)  | 1         | 2                                                                                                                                                                                                                                                                                                                                                                                   | 3         | 5    |   |   |   |            |     |     |     |      |
| y (output) | 4.8       | 7.2                                                                                                                                                                                                                                                                                                                                                                                 | 8.6       | 12.8 |   |   |   |            |     |     |     |      |
| Q3         | a)        | Suppose you train a logistic regression classifier $h_{\theta}(x)=g(\theta_0+ \theta_1x_1+ \theta_2x_2 + \theta_3x_3)$ , where $g$ : sigmoid function, $\theta_0=-6$ , $\theta_1=2$ , $\theta_2=3$ , $\theta_3=5$ . By applying classifier $h_{\theta}(x)$ on the data points: A (1,1, 1), B(-1,1,-1), C(2,1, 1), D(4,-1,0), mention which data points belong to the same category. | 5 M       |      |   |   |   |            |     |     |     |      |
|            | b)        | Explain in detail the working of the Hard margin SVM Classifier                                                                                                                                                                                                                                                                                                                     | 5M        |      |   |   |   |            |     |     |     |      |
| Q4         | a)        | Draw the Decision Tree for the following Boolean functions                                                                                                                                                                                                                                                                                                                          | 5 M       |      |   |   |   |            |     |     |     |      |
|            | i)<br>ii) | $\rightarrow (A \wedge \neg B) \rightarrow T / F$ $A \vee [B \wedge C]$ <p style="text-align: right;"><i>ambiguity</i></p>                                                                                                                                                                                                                                                          |           |      |   |   |   |            |     |     |     |      |



# DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY

(Deemed to be University), Girinagar, Pune - 411025

M.Tech/PhD EXAMINATIONS

END SEMESTER EXAMINATION (SPRING:2023)

Course(Subject): Computational Number Theory & Cryptography

Course Code: AM 628 / 628D

Maximum Marks: 50

Duration : 03 hrs

- Please ensure that you have received the correct question paper.
- Marks are indicated against each question.
- Write clearly and legibly.
- Answer all 5 questions. This Question paper contains two pages

1. (a) Find the last digit in the decimal value of  $1997^{1998^{1999}}$   
 (b) By using Modular Exponentiation(M.E.), find the units digit in the decimal value of  $1077^{1777^{1277^{1377}}}$   
 (c) Define a *Fermat's* little theorem. By using this, find the remainder when  $15^{1976}$  is divided by 23  
 (d) Define the Legendre Symbol. By using the Law of Quadratic Reciprocity, Compute (i) (-1776 / 1013) (ii) (1428 / 2411). (4X2.5=10)
2. (a) By using Chinese Remainder Theorem(CRT), Solve the following linear system,  $x \equiv 2(\text{mod } 4)$ ,  $x \equiv 3(\text{mod } 5)$ ,  $x \equiv 4(\text{mod } 9)$ ,  $x \equiv 5(\text{mod } 13)$ .  
 (b) Find a Factor of 31861, using the Pollard rho method, the given seed value  $x_0 = 1$ , and the given polynomial  $f(x) = x^2+1$ . (5+5)
3. (a) Define the Jacobi symbol. Using the generalized law of quadratic reciprocity, the following quadratic congruences are solvable or not. If so solve it.  
 i.  $x^2 \equiv 442(\text{mod } 385)$   
 ii.  $x^2 \equiv -198(\text{mod } 2873)$  (4)  
 (b) Define Caesar cipher. Decipher each ciphertext created by the Caesar cipher.  
 i. QHFHV VLWBL VWKHP RWKHU RILQY HQWLR Q  
 ii. KDYHD QLFHG DB (6)
4. Using the exponentiation modulus  $P = 3037$  and the enciphering key  $c = 31$ , decipher the ciphertext 0624 1435 2669 0998 (10)



**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
**(Deemed to be University), Girinagar, Pune - 411025**  
**M.Tech/PhD EXAMINATIONS**  
**THIRD MID SEMESTER EXAMINATION (SPRING:2023)**

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Course(Subject) Name: **Computational Number Theory & Cryptography**

Course Code: **AM 628 / AM 628D**

Duration : **01 Hr(s)**

Maximum Marks: **10**

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- Please ensure that you have the correct question paper.
  - Marks are indicated against each question.
  - Write clearly and legibly.
  - **All Questions are compulsory.**
- 

- |                                                                                                       |      |
|-------------------------------------------------------------------------------------------------------|------|
| 1. State Wilson's theorem                                                                             | (1M) |
| 2. State Euler's theorem                                                                              | (1M) |
| 3. Compute Legendre Symbol (1428/2411)                                                                | (1M) |
| 4. Compute $\varphi(n)$ when $n = 183920$                                                             | (1M) |
| 5. Compute (152/43) and (3797/7297)                                                                   | (2M) |
| 6. Define a quadratic residue and hence determine whether 10 and 7 are quadratic residues of 13.      | (2M) |
| 7. State Fermat's Little theorem and hence determine the remainder when $24^{1947}$ is divided by 17. | (2M) |

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
 (Deemed to be University), Girinagar, Pune - 411025  
**M.Tech/PhD EXAMINATIONS**  
**THIRD MID SEMESTER EXAMINATION (SPRING:2023)**

Course(Subject) Name: **Computational Number Theory & Cryptography**

Course Code: **AM 628 / AM 628D**

Duration : **01 Hr(s)**

Maximum Marks: **10**

- Please ensure that you have the correct question paper.
- Marks are indicated against each question.
- Write clearly and legibly.
- **All Questions are compulsory.**

1. State Wilson's theorem (1M)
2. State Euler's theorem (1M)
3. Compute Legendre Symbol  $(1428/2411)$  (1M)
4. Compute  $\varphi(n)$  when  $n = 183920$  (1M)
5. Compute  $(152/43)$  and  $(3797/7297)$  (2M)
6. Define a quadratic residue and hence determine whether 10 and 7 are quadratic residues of 13. (2M)
7. State Fermat's Little theorem and hence determine the remainder when  $24^{1947}$  is divided by 17. (2M)

$$(-1)^8$$

$$(-1)^{\frac{13}{2}}$$

$$(-1)^{\frac{13}{2}}$$

$$2^{\frac{2-1}{2}}$$

$$(-1)^{\frac{13}{2}}$$

$$2 \cdot (-1)^{\frac{1}{2}}$$

$$11^{\frac{10}{11}}$$

$$\frac{2410}{2}$$

$$(-1)^{(1)(3)}$$

$$(-1)^1$$

$$(-1)^{3 \times}$$

$$2^{\frac{3-1}{2}}$$

$$2^1$$

$$(-1)^{\frac{1}{2}(\frac{1}{2})}$$

$$1$$

$$121$$

$$100$$

$$9 \ 9 \ 9$$

$$2^2 \equiv$$

$$\left(\frac{5}{23}\right) = (-1)^{\frac{23-1}{2}}$$

$$3^2 \equiv 1 \pmod{p}$$

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
(Deemed to be University), Girinagar, Pune - 411025  
**M.Tech/PhD EXAMINATIONS**  
**FIRST MID SEMESTER EXAMINATION (SPRING:2023)**

---

Course(Subject): **Computational Number Theory & Cryptography**

Course Code: **AM 628 /AM 628D**

Maximum Marks: **10**

Duration: **01 hrs**

---

- Please ensure that you have the correct question paper.
  - Marks are indicated against each question.
  - Write clearly and legibly.
- 

1. Define (i) Cryptography (ii) Plaintext and Ciphertext (iii) Cipher encryption and Decryption (iv) Fermat Number( $f_n$ ). (2M)
2. What are the different extensions of Number theory? Briefly explain about each extension of its importance in cryptography. (2M)
3. By using Floor or Ceiling function, find an equation(relation) that the number of leap years ' $\ell$ ' after 1600 and not exceeding a given year ' $y$ '. (2M)
4. Define Fundamental Theorem of Arithmetic(F.T.A). (1M)  
By using F.T.A and Floor or Ceiling function
  - (a) Find the canonical decomposition of the composite number  $2^{27}+1$ . (1.5M)
  - (b) Find the number of trailing zeros in the decimal value of  $609!$  (1.5M)

**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
 (Deemed to be University), Girinagar, Pune - 411025  
**M.Tech/PhD EXAMINATIONS**  
**SECOND MID SEMESTER EXAMINATION (SPRING:2023)**

Course(Subject): **Computational Number Theory & Cryptography**

Course Code: **AM 628 & AM 628D**

Maximum Marks: **10**

Duration: **01 hrs**

- Please ensure that you have the correct question paper.
- Marks are indicated against each question.
- Write clearly and legibly.

1. The International Standard Serial Number (ISSN) is an internationally accepted code for identifying serial publications. It consists of two four-digit groups. The eighth digit  $d_8$ , which can be an X (for 10), is a check digit, defined by  $d_8 \equiv (d_1, d_2, \dots, d_7) \cdot (8, 7, 6, 5, 4, 3, 2) \pmod{11}$ . Compute / Verify the check digit for each seven-digit identification number 0 - 07 - 095831 - 2. (2M)
2. Compute the missing check digit in each vehicle identification number mentioned below:  
 2HGES165-1H541873 87654321098765432 (2M)
3. Airlines assign a check digit 'd' to their 11-digit ticket numbers  $d_1, d_2, \dots, d_{11}$ , defined by  $d \equiv d_1, d_2, \dots, d_{11} \pmod{7}$ . (In fact, the ticket number contains 14 digits and the check digit, but the three-digit airline code is not entered into the calculation of the check digit.) Compute the check digit if the 11-digit ticket number is 20754376405. (2M)
4. What is Pollard Rho Method. Find a Factor of 39997, using the Pollard rho method, the given seed value  $x_0 = 1$ , and the given polynomial  $f(x) = x^2 + 1$ . 23 × 3747 (4M)

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8 7 6 5 4 3 2 1 0 9 8 7 6 5 4 3 2

8 7 6 5 4 3 2 1 0 8 7 6 5 4 3 2

8 7 6 5 4 3 2



**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
**(Deemed University)**  
**MTech/MS/PhD EXAMINATIONS**  
**END SEMESTER EXAMINATION – (SPRING/AUTUMN 2023)**

Course (Subject): Computer Vision

Code: CE-632

Duration : 03 HOURS

Max Marks: 50

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Note:

1. Answer **ALL** questions.
  2. Marks are indicated against each Question.
  3. Question paper consists of: 01 Pages.
- 

**Q1.** **[10 Marks]**  
Explain Circular Hough Transform in detail? Can we use it for boundary detection? For which kind of applications, Hough Transform is more suitable?

**Q2.** **[10 Marks]**  
What do you mean by Morphological operations? Explain opening and closing Morphological operations in detail. Is Morphological operation same as filtering operation? In what kind of applications, Morphological operations are suitable?

**Q3.** **[10 Marks]**  
Write a short note scale-invariant feature transform (SIFT) feature extraction method.

**Q4.** **[10 Marks]**  
Explain working of Canny Edge Detection Algorithm. Can it be used to detect corners? If no, which methods are suitable for corner detection?

**Q5.** **[10 Marks]**  
Explain HoG (Histogram of Gradients) feature descriptor. How it is calculated? For which applications, it can be used?

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**DEFENCE INSTITUTE OF ADVANCED TECHNOLOGY**  
**(Deemed University)**

MTech/MS/PhD EXAMINATIONS

Quiz-II EXAMINATION – (SPRING/AUTUMN 2023)

Course (Subject): Computer Vision

Code: CE-632

Duration : 01 HOURS

Max Marks: 10

Date: 21/03/2022 Time: 12 Hrs.-13 Hrs.

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Note:

1. Answer **ALL** questions.
  2. Marks are indicated against each Question.
  3. Question paper consists of: 01 Pages.
- 

**Q1.**

**[5 Marks]**

Explain working of Canny Edge Detection Algorithm.

**Q2.**

**[5 Marks]**

Explain Hough Transform for circle detection in detail.

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MTech/MS/PhD EXAMINATIONS

Quiz-III EXAMINATION – (SPRING/AUTUMN 2023)

Course (Subject): Computer Vision

Code: CE-632

Duration : 01 HOURS

Max Marks: 10

Date: 25/04/2022 Time: 12 Hrs.-13 Hrs.

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Note:

1. Answer **ALL** questions.
  2. Marks are indicated against each Question.
  3. Question paper consists of: 01 Pages.
- 

**Q1.**

Explain working of any Corner Detector.

**[5 Marks]**

**Q2.**

Explain Boundary tracking procedure in detail.

**[5 Marks]**

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MTech/MS/PhD EXAMINATIONS

Quiz-I EXAMINATION – (SPRING/AUTUMN 2023)

Course (Subject): Computer Vision

Code: CE-632

Duration : 01 HOURS

Max Marks: 10

Date: 21/02/2022 Time: 12.00-13.00Hrs

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Note:

1. Answer **ALL** questions.
  2. Marks are indicated against each Question.
  3. Question paper consists of: 01 Pages.
- 

**Q1.**

**[5 Marks]**

Explain any Image Thresholding technique in detail.

**Q2.**

**[5 Marks]**

Write a short note on connected component labelling algorithm.

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# PGC 602 – Stress Management by Yoga

Duration - 120 mins

2022-24 Semester II

Max Marks - 50

## Section A - Multiple Choice Question

(1 Mark x 10 = 10 Marks)

1. The energy responsible for inhalation and the intake of air, as well as the functioning of the heart and lungs.
  - a. ApanaVayu
  - ☒ b. PranaVayu
  - c. VyanaVayu
  - d. UdanaVayu
2. Which phase of Breathing is known as Rechaka?
  - a. Inhalation
  - b. Exhalation
  - c. Internal Breath Retention
  - d. External Breath Retention
3. What is a stressor?
  - a. A person experiencing stress
  - b. An external event or situation that triggers a stress response
  - c. A hormone released during stressful situations
  - d. A relaxation technique for managing stress
4. Patanjali Yoga Sutras comprises of how many chapters
  - a. 2
  - b. 5
  - c. 3
  - d. 4
5. What are the Pillars of Balanced Living
  - ☒ a. Aachar, Vichar, Ahar, Vihar
  - b. Avidya, Asmita, Raga, Dvesha
  - c. Yama, Niyama, Asana, Pranayama
  - d. Ahimsa, Satya, Asteya, Bhramcharya
6. Which of the following is an example of a healthy coping mechanism for stress?
  - a. Smoking cigarettes
  - b. Drinking alcohol
  - ☒ c. Engaging in regular physical exercise
  - d. Overeating
7. What is the term used to describe a technique that involves focusing attention on the present moment to reduce stress and enhance well-being?
  - ☒ a. Meditation
  - b. Visualization
  - c. Yoga
  - d. Cognitive restructuring
8. Which of the following statements about stress is true?
  - a. Stress is always harmful and should be avoided at all costs.
  - ☒ b. Stress is solely caused by external factors and cannot be managed internally.
  - ☒ c. Stress can have both positive and negative effects on individuals.
  - d. Stress is a rare occurrence and does not affect most people.